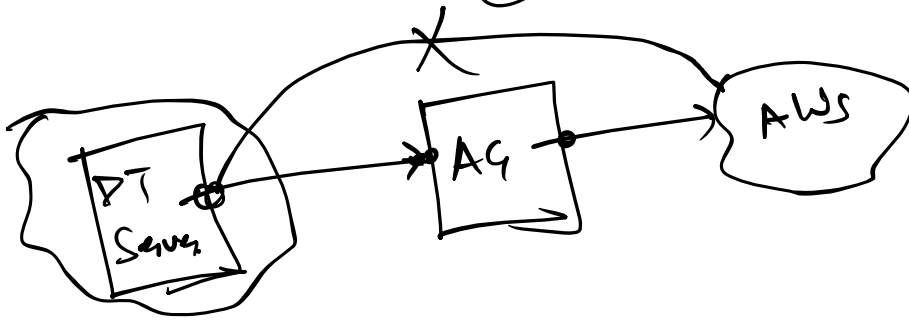


- ① Cloud Monitoring.
- ② K8s cluster monitoring.
- ③ user Mng. & Access Control.

- ① Active Gate
- ① Private Synthetic Monitoring
- ② cloud monitoring.
- ③ External DB Monitoring.



- ① Linux Server
- ② Active Gate.
- ③ Connect AG with DT Server (CloudFormation)
- ④ cloud monitoring → Stack

AWS CloudFormation - IaC (Infrastructure as Code)

↓

Terraform.

create, manage, and Automate AWS resources with template.

↳ yaml or json

Why?

- ① Automate AWS infra. setup
- ② Avoid Manual error.

- ①
- ② Avoid Manual error.
- ③ Make info. reusable.
- ④ Easy to version control.
- ⑤ Support rollback if deployment fails.

Template - yaml / json. - Blueprint

Stack - A deployed instance of the Template. - Actual Code.

Resource - Actual AWS Services.

Parameters - Input given during deployment.

Output - Useful values returned.

Basic flow:

- ① Write template
- ② upload into cloud front.
- ③ Create stack
- ④ Provision resources
- ⑤ update stack later
- ⑥ Delete stack to remove resources.

K8s cluster Monitoring:

① K8s

② cluster Architecture

(1) K8s

(2) cluster Archiecture

(3) Connection

(4) Details

(1) K8S:- Open Source ^{code is free} container ^{→ Docker} orchestration Platform ^{→ system/tool where all will run}
used to deploy, manage (4 scale appli' automatically.
Multiple Container automatically

(1) Code

(2) Libraries

(3) Dependencies

Solves:-

(1) Auto Deployment.

(2) Auto scaling.

(3) Self healing.

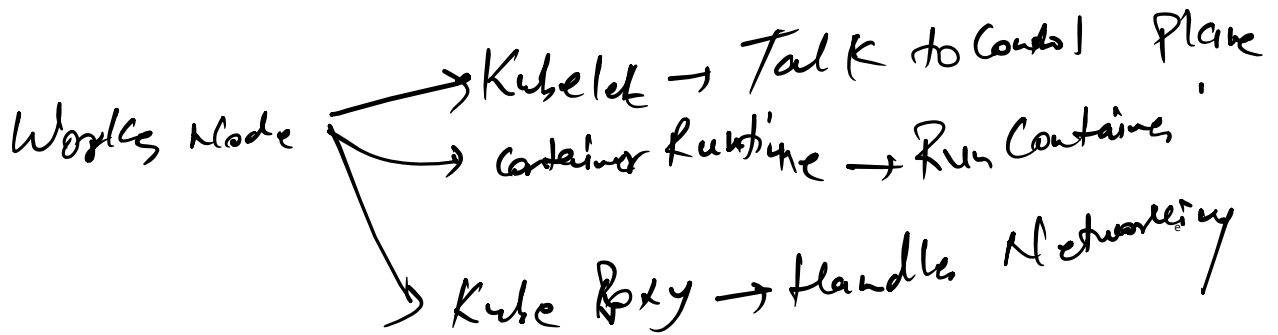
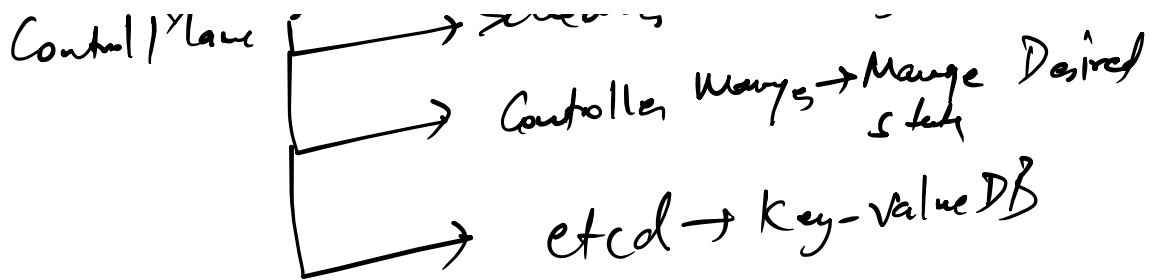
(4) Load Balancing.

Core Archiecture:-

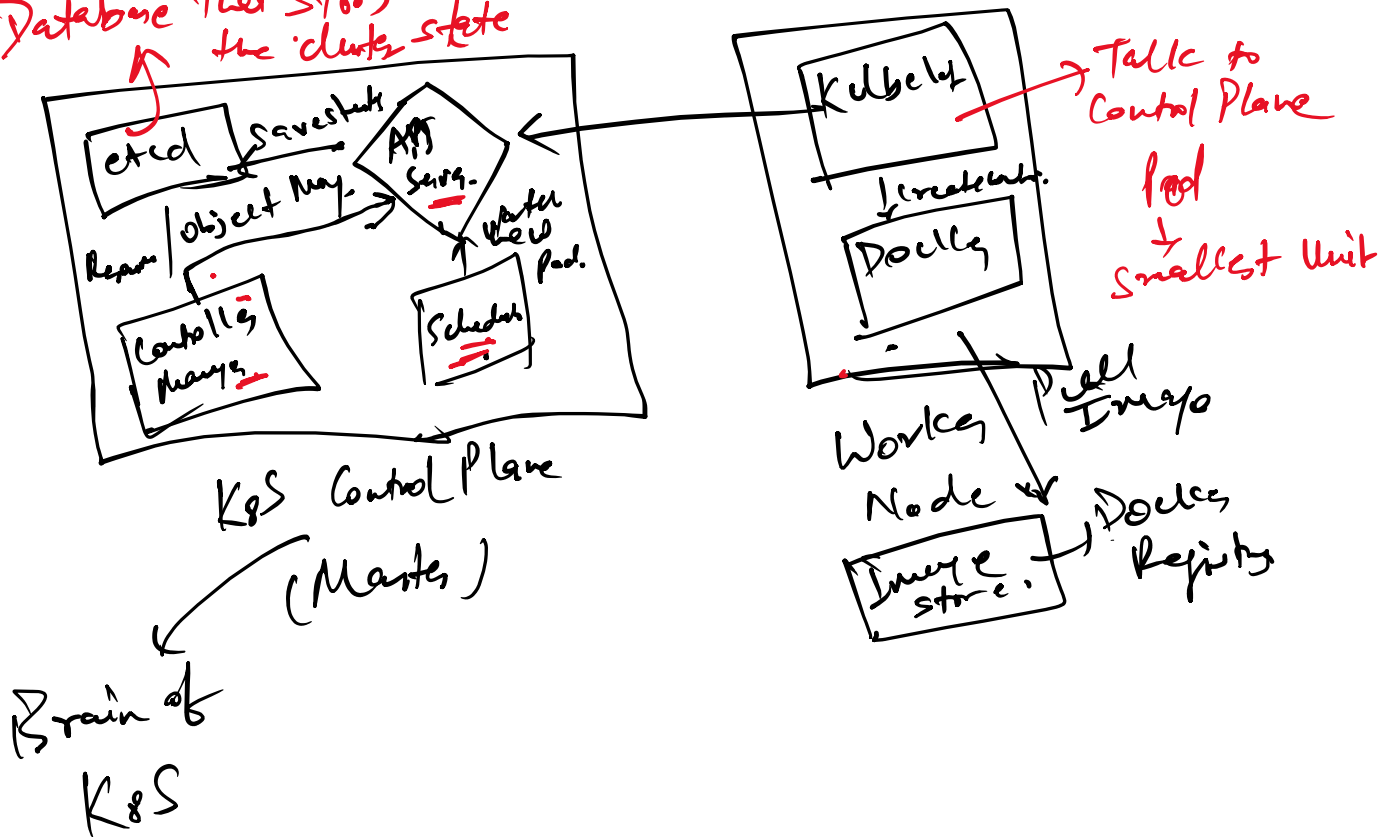
(1) Control Plane

(2) Worker Node.

Control Plane
 → API Server → entry point
 → Scheduler → Assign Pod to Nodes
 → Controller Manager → Manage Desired



Database that stores the cluster state



① Kubelet apply -f app.yaml.

↓

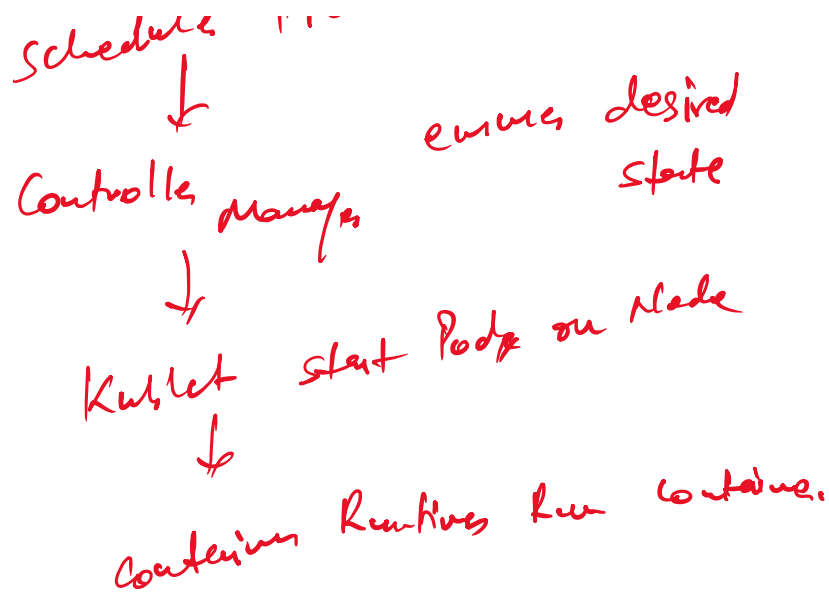
API Serv. receive request

↓

Scheduler Picks Best Node

↓

... desired



③ User Mng. & Access Control

- ① User → Add users, edit them, Assign group
- ② Group → Assign Permission Group wise.
- ③ Permission | Role $\begin{cases} \rightarrow \text{Account level} \\ \rightarrow \text{Env. level permission} \end{cases}$
- ④ IAM →
- ⑤ SSO → Group mapping.

Admin group → full platform.
 Operation group → Monitor DB, Problem Analysis
 Developer group → App observability Access only
 Auditor / Views group → read-only Access.